

Why does the construction give a basis of S ?

Reverse the construction: how can we check n given solutions?

Check whether $\Phi(y_1), \dots, \Phi(y_n)$ are _____.

2. The Wronskian test

For n functions $y_1, \dots, y_n$, define their **Wronskian** by

$$W(t) = \det \begin{pmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ \dot{y}_1(t) & \dot{y}_2(t) & \cdots & \dot{y}_n(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{pmatrix}.$$

At $t = t_0$, its columns are the vectors $\Phi(y_1), \dots, \Phi(y_n)$.

Procedure for checking a fundamental set

- (a) Check that $y_1, \dots, y_n$ are _____.
- (b) Pick an arbitrary $t_0 \in I$.
- (c) The functions form an FSS if and only if _____.

Remark. For solutions of (1), if $W(t_0) \neq 0$ for some $t_0 \in I$, then _____ for every $t \in I$.

Example 11.1 – Checking a fundamental set. Consider

$$\ddot{y} + 2\dot{y} + y = 0.$$

We claim that $y_1(t) = e^{-t}$ and $y_2(t) = te^{-t}$ form an FSS on $I = \mathbb{R}$.

- (a) Check that both functions are solutions of the ODE.

- (b) Choose $t_0 = 0$ and evaluate the Wronskian.

(c) **State the conclusion.**

Therefore $\{e^{-t}, te^{-t}\}$ is a _____ on $\mathbb{R}$.

3. Nonhomogeneous linear ODEs

Consider

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)\dot{y} + a_0(t)y = g(t). \quad (2)$$

Assume that $a_0, \dots, a_{n-1}$ and g are continuous on $I = (a, b)$. Suppose y_p is a particular solution of (2), and $\{y_1, \dots, y_n\}$ is a fundamental set of solutions (FSS) of the associated homogeneous equation.

Theorem 11.2 (Structure of the general solution). The general solution of (2) has the form

$$y(t) = y_p(t) + \underbrace{\hspace{10em}}_{y_c(t)}, \quad c_i \in \mathbb{R}. \quad (3)$$

The constants c_i are determined by the _____.

How do the initial conditions change for the homogeneous part?

For the IVP with $y^{(j)}(t_0) = \eta_j$, $j = 0, \dots, n-1$, and $t_0 \in I$, the solution y_c of the homogeneous equation must satisfy

$$y_c^{(j)}(t_0) = \underline{\hspace{10em}}, \quad j = 0, \dots, n-1.$$

Direct verification.

Verify that $y_p + y_c$ solves (2). Conversely, if y solves (2), explain why $y - y_p$ is the solution of the homogeneous ODE $x^{(n)} + a_{n-1}(t)x^{(n-1)} + \cdots + a_1(t)\dot{x} + a_0(t)x = 0$ with initial conditions $x^{(j)}(t_0) = \eta_j - y_p^{(j)}(t_0)$.

Procedure for solving a linear ODE

- (a) Find an FSS for the _____.
- (b) Find a _____ of (2).
- (c) Set $y(t) = y_p(t) + \sum_{i=1}^n c_i y_i(t)$.
- (d) Impose the initial conditions and solve for _____.

4. Initial conditions and constant coefficients

Write the system that determines the constants.

With $\mathbf{c} = (c_1, \dots, c_n)^\top$, define

$$M_0 = \begin{pmatrix} y_1(t_0) & \cdots & y_n(t_0) \\ \dot{y}_1(t_0) & \cdots & \dot{y}_n(t_0) \\ \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t_0) & \cdots & y_n^{(n-1)}(t_0) \end{pmatrix}, \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_0 \\ \eta_1 \\ \vdots \\ \eta_{n-1} \end{pmatrix}, \quad \mathbf{p}_0 = \begin{pmatrix} y_p(t_0) \\ \dot{y}_p(t_0) \\ \vdots \\ y_p^{(n-1)}(t_0) \end{pmatrix}.$$

Imposing the initial conditions in (3) gives

$$M_0 \mathbf{c} = \underline{\hspace{2cm}}, \quad \mathbf{c} = \underline{\hspace{2cm}}.$$

Remark. M_0 is invertible because its columns are the initial vectors of an FSS.

Linear homogeneous constant-coefficient ODEs

We now consider

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1\dot{y} + a_0y = 0, \quad a_0, \dots, a_{n-1} \in \mathbb{R}. \quad (4)$$

For variable coefficients, explicit solutions are generally difficult to find; integrating factors and reduction of order apply in suitable cases. For constant coefficients, an FSS can be found systematically, Laplace transforms can give particular solutions in many cases, and stability can be tested directly. Here we study the cases $n = 1$ and $n = 2$.

5. Case $n = 1$. Solve and classify the zero equilibrium.

Write $\dot{y} + a_0y = 0$ as $\dot{y} = \lambda y$, where $\lambda = -a_0$. An FSS and the general solution are

$$\text{FSS} = \underline{\hspace{2cm}}, \quad y(t) = \underline{\hspace{2cm}}.$$

Sketch $y(t)$ for the three signs of λ , with the same $y(0) > 0$.



Condition	Stability of $y = 0$
$\lambda > 0$	
$\lambda = 0$	
$\lambda < 0$	

6. Second-order equations: distinct real roots

Consider

$$\ddot{y} + a_1\dot{y} + a_0y = 0. \quad (5)$$

Test an exponential solution $y(t) = e^{\lambda t}$.

Substitute into (5) and use $e^{\lambda t} \neq 0$ to obtain an informative equation about λ .

The **characteristic polynomial** is $p(\lambda) = \underline{\hspace{4cm}}$. Its roots give exponential solutions. Define its discriminant by

$$\Delta = \underline{\hspace{4cm}}.$$

Case I: $\Delta > 0$ (distinct real roots).

The roots and corresponding solutions are

$$\lambda_{1,2} = \underline{\hspace{4cm}}, \quad y_i(t) = e^{\lambda_i t}, \quad i = 1, 2.$$

Are these solutions an FSS? Compute the Wronskian at $t_0 = 0$.

The general solution is

$$y(t) = \underline{\hspace{4cm}}.$$

Classify the zero equilibrium in state space.

The state is $x = (y, \dot{y})^T$ and the zero equilibrium is $\bar{x} = (0, 0)^T$.

Root condition	Stability of $\bar{x} = 0$
$\lambda_1 < 0$ and $\lambda_2 < 0$	
At least one root is positive	
One root is negative; the other is zero	

7. Second-order equations: a repeated real root**Case II:** $\Delta = 0$.

The repeated root is $\lambda_1 = \lambda_2 = \lambda$, where $\lambda =$ _____. The characteristic polynomial gives one solution $y_1(t) = e^{\lambda t}$.

Check that $y_2(t) = te^{\lambda t}$ is a second solution.

Compute its first two derivatives and substitute into (5).

Verify that $\{e^{\lambda t}, te^{\lambda t}\}$ is an FSS.

The general solution is

$$y(t) = \underline{\hspace{10cm}}.$$

Classify $\bar{x} = (0, 0)^\top$ for each sign of λ .

Condition	Stability of $\bar{x} = 0$
$\lambda < 0$	
$\lambda = 0$	
$\lambda > 0$	

Explain carefully why the repeated root $\lambda = 0$ gives instability.

Complete the state trajectory:

$$x(t) = \begin{pmatrix} y(t) \\ \dot{y}(t) \end{pmatrix} = \underline{\hspace{10cm}}.$$

8. Second-order equations: complex conjugate roots**Case III: $\Delta < 0$.**

Write

$$\lambda_{1,2} = -\frac{a_1}{2} \pm i \frac{\sqrt{-\Delta}}{2} = a \pm ib, \quad a = -\frac{a_1}{2}, \quad b = \frac{\sqrt{-\Delta}}{2} > 0.$$

The two complex solutions are

$$\hat{y}_{1,2}(t) = e^{(a \pm ib)t} = e^{at}(\cos(bt) \pm i \sin(bt)).$$

Both their real and imaginary parts are real solutions. Thus take

$$y_1(t) = \underline{\hspace{4cm}}, \quad y_2(t) = \underline{\hspace{4cm}}.$$

Is this an FSS? Compute the Wronskian at $t_0 = 0$.

The real general solution is

$$y(t) = \underline{\hspace{10cm}}.$$

Classify the zero equilibrium using the real part a .

Condition	Stability of $\bar{x} = (0, 0)^T$
$a = 0$	
$a < 0$	
$a > 0$	

Sketch nearby orbits in the $(y, \dot{y})$ plane. Indicate direction.